

$n = p^2$
 $a_1, \dots, a_p$
 $a_1, \dots, a_p$

Indian Statistical Institute
 Mid-semestral Examination: 2024-25
 Programme: Master of Mathematics
 Course: Number Theory

Maximum Marks: 40

Duration: 2 Hours

Date: February 25, 2025

Instruction: Answer all questions. Write brief answers in complete sentences. You need to explain all the important steps in order to obtain full marks.

1. Suppose $n, d > 1$ and $d|n$. Show that the natural reduction map $f: U(n) \rightarrow U(d)$ is surjective. [8]

2. Suppose $a_1, a_2, \dots, a_p$ are integers that are pairwise distinct modulo p for a prime $p > 2$ and similarly $b_1, b_2, \dots, b_p$ are integers that are pairwise distinct modulo p . Show that $a_1b_1, a_2b_2, \dots, a_pb_p$ cannot be pairwise distinct modulo p . [12]

3. Suppose

$$c_q(n) := \sum_{a \in U(q)} e\left(\frac{na}{q}\right).$$

Show that

$$c_q(n) = \sum_{d|(q,n)} d\mu(q/d).$$

[10]

4. Let $q > 1$ be an integer. Let V_q be the $\mathbb{C}$ -vector space of q -periodic arithmetic functions; i.e.,

$$V_q = \{f: \mathbb{Z} \rightarrow \mathbb{C} : f(n+q) = f(n) \text{ for all } n \in \mathbb{Z}\}.$$

(a) Give a basis for V_q (No proof necessary). [2]

(b) Let us define a linear operator $F: V_q \rightarrow V_q$ by

$$F(f)(x) := \frac{1}{\sqrt{q}} \sum_{y \pmod{q}} f(y) e\left(\frac{xy}{q}\right).$$

Determine $F(F(f))$ in terms of f . [5]

(c) Use part (b) to show that F is an isomorphism and write the inverse of F explicitly. [2]

(d) Write a list of complex numbers that can possibly be the eigenvalues of F . Justify your answer. [4]

$\sum_{d|n} \varphi(d) = n$
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$$\begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$